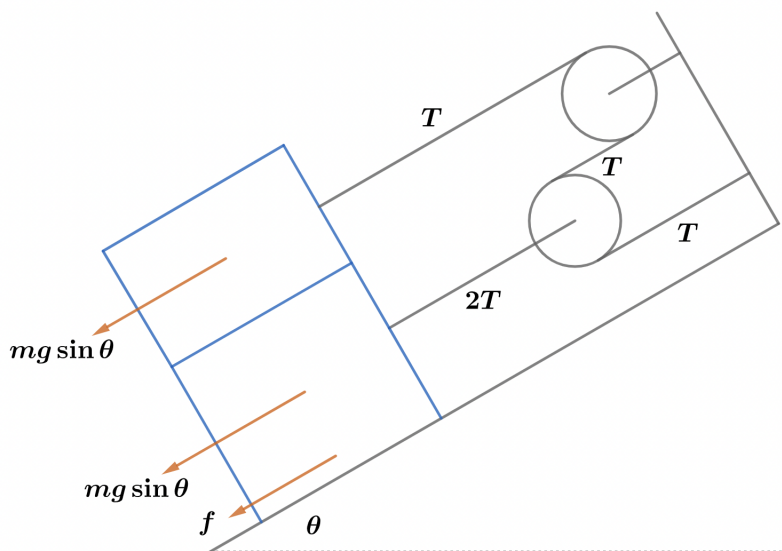


2022B F=ma Exam: Problem 12

Kevin S. Huang



Since the tension on the lower block is twice the tension on the upper block, friction must point downwards along the plane to oppose relative motion. Balancing forces on the upper block,

$$T = mg \sin \theta$$

Balancing forces on the lower block,

$$mg \sin \theta + f = 2T = 2mg \sin \theta$$

$$f = mg \sin \theta$$

For static friction, $f \leq \mu N$. The normal force is $N = 2mg \cos \theta$ so we have

$$mg \sin \theta \leq 2\mu mg \cos \theta$$

$$\mu \geq \frac{\tan \theta}{2}$$

and the answer is A.